

Secondary Storage

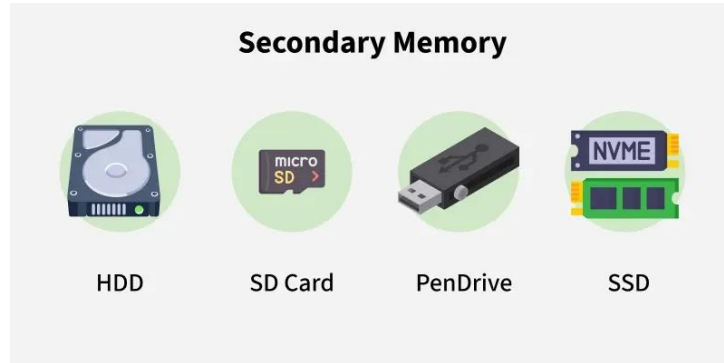
Major Types of Secondary Storage +
Structure of HDD, SSD

Secondary Storage

Secondary memory, also known as **secondary storage**, refers to the storage devices and systems used to store data persistently, even when the computer is powered off.

Unlike **primary memory** (RAM), which is **fast and temporary**, **secondary memory** is slower but offers much larger storage capacities.

Some Examples of secondary memory include **hard disk drives (HDDS)**, **solid-state drives (SSDs)**, **optical disks (CDS/DVDS)**, and **external storage devices like USB drives**.



Use of Secondary Memory

Secondary memory is used for different purposes, but the main purposes of using secondary memory are:

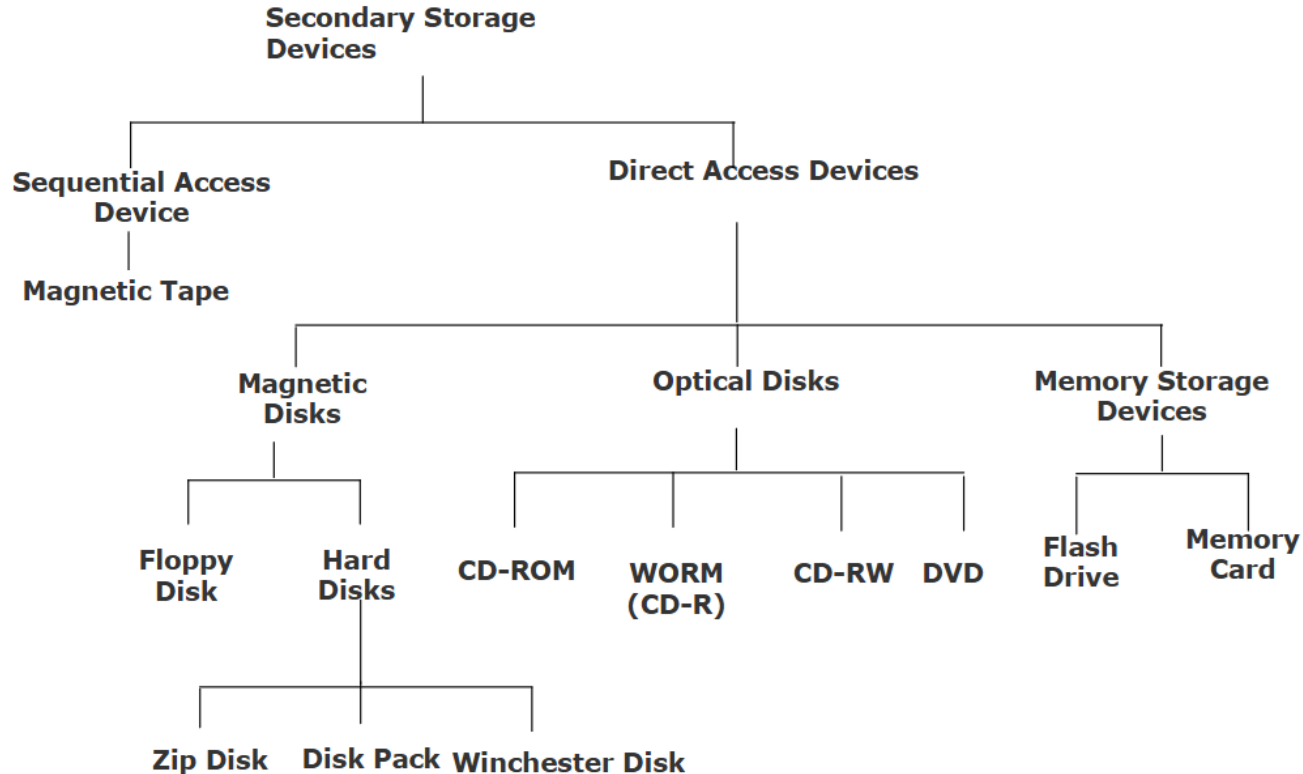
- **Permanent storage:** As we know that primary memory stores data only when the power supply is on, it loses data when the power is off. So we need a secondary memory to store data permanently, even if the power supply is off.
- **Large Storage:** Secondary memory provides large storage space so that we can store large data like videos, images, audios, files, etc, permanently.
- **Portable:** Some secondary devices are removable. So we can easily store or transfer data from one computer or device to another.

These devices are essential for long-term data storage and retrieval, providing a means to store operating systems, applications, and personal files, ensuring that data remains intact even after the system is turned off.

Features of Secondary Storage:

- Used to hold stored program instructions
- Used to hold data and information of stored jobs
- Lot cheaper than primary storage
- Retains data even without power
- Has an operating speed far slower than that of the primary storage
- Used to store large volumes of data on a permanent basis
- Also known as auxiliary memory

Classification of Common Secondary Storage Devices



Sequential or Serial Access Storage Device

- Arrival at the desired storage location may be preceded by sequencing through other locations
- Data can only be **retrieved in the same sequence in which it is stored**
- Access time varies according to the storage location of the information being accessed
- Sequential access means that computer must run through the data in sequence, starting from the beginning, in order to locate a particular piece of data.
- Suitable for sequential processing applications where most, if not all, of the data records need to be processed one after another

Eg: Magnetic tape is an example of sequential access device. Let us suppose that magnetic tape consists of 80 records. To access the 25th record, the computer starts from first record, then reaches second, third etc. until it reaches the 25th record.

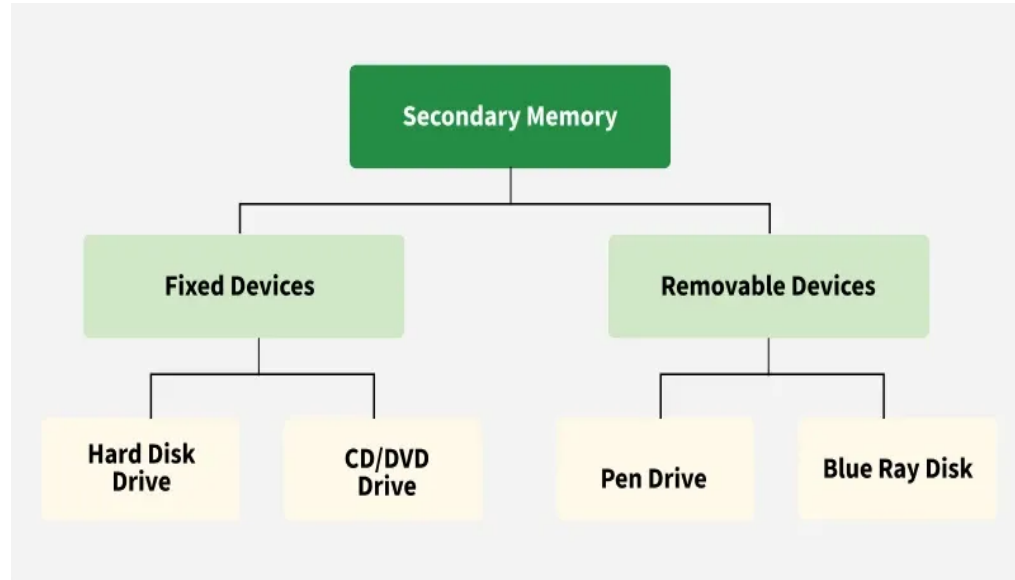
Sequential access devices are generally slow devices.

Direct or Random Access Storage Devices

- Direct access devices are the ones in which **any piece of data can be retrieved** in a **non-sequential manner by locating it using the data's address**.
- It accesses the data directly, from a desired location.
- There is **no predefined order** in which one can read and write data from a direct access device.
- Magnetic, optical are typical examples of such a storage device

Eg: In a magnetic disk consisting of 80 records, to access the 25th record, the computer can directly access the 25th record, without going past the first 24 records.

Types of Secondary Memory



1. Fixed Devices

Fixed devices in secondary memory are storage devices that are **permanently installed in a system and cannot be easily removed**, like internal **hard drives or solid-state drives (SSDs)**.

They **store data that is always accessible by the system**.

Hard Disk Drive (HDD)

- A **Hard Disk Drive (HDD)** is a traditional storage device that stores data on spinning magnetic disks. It is commonly used because it offers large storage space at a low cost. However, it is slower than newer storage technologies.

Solid-State Drive (SSD)

- A **Solid-State Drive (SSD)** is a faster and newer type of storage. It uses flash memory instead of spinning disks, so it has no moving parts. This makes it more reliable and much faster for reading and writing data compared to an HDD.

2. Removable Devices

Removable devices in secondary memory are storage devices that can be easily disconnected and used on different systems, like USB drives or external hard drives.

They **allow for easy data transfer and backup.**

Optical Discs (CD, DVD, Blu-ray)

- **CD (Compact Disc):** Holds up to 700 MB of data, often used for music, software, or small files.
- **DVD (Digital Versatile Disc):** Can store more data than a CD, typically 4.7 GB or more, and is commonly used for videos or larger data files.
- **Blu-ray Disc:** Designed for high-definition video, Blu-ray discs can hold from 25 GB (single layer) to 50 GB (dual layer), making them great for movies and large files.

USB Flash Drives

- **USB Flash Drives** are small, portable devices that use flash memory to store data. They're commonly used to transfer files between computers or as backup storage. They are durable, easy to carry, and come in various sizes, ranging from a few gigabytes to several terabytes.

Magnetic Tapes

- Magnetic Tapes are an older form of storage where data is stored on long, thin tapes. While they are not commonly used for personal computers anymore, they are still used for large-scale data storage and archiving because they offer a lot of space at a low cost.

Flash Memory Cards (SD Cards, MicroSD Cards)

- **Flash Memory Cards** like **SD cards** and **MicroSD cards** are tiny, portable storage devices used in cameras, smartphones, and other gadgets. They are ideal for storing photos, videos, and other media files.

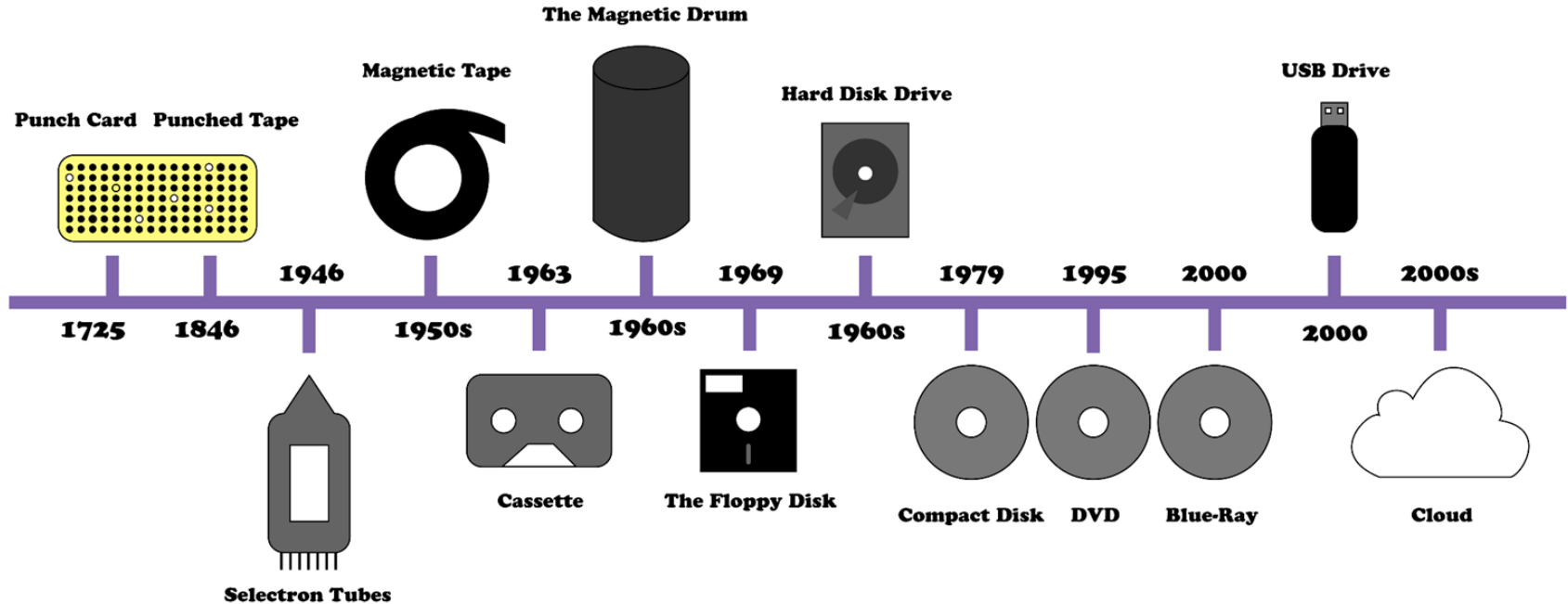
External Hard Drives (If used as a removable device)

- While external hard drives can be considered fixed if connected permanently, they are usually removable and used for backup or transferring large files.

Cloud Storage

- **Cloud Storage** is not a physical device but a service that allows you to store your data online, on servers that you can access over the internet. Popular services like **Google Drive**, **Dropbox**, and **OneDrive** allow users to store, share, and access their data from anywhere with an internet connection.

Data Storage Evolution Timeline (approx)



MAGNETIC TAPE:



Magnetic tape is a plastic tape with magnetic coating.

It is a storage medium on a large open reel or in a smaller cartridge or cassette (like a music cassette).

Magnetic tapes are cheaper storage media.

They are durable, can be written, erased, and re-written.

Magnetic tapes are sequential access devices, which mean that the tape needs to rewind or move forward to the location where the requested data is positioned in the magnetic tape.

Due to their sequential nature, magnetic tapes are not suitable for data files that need to be revised or updated often.

They are generally used to store back-up data that is not frequently used or to transfer data from one system to other.

MAGNETIC DISK

Magnetic disk is a direct access secondary storage device.

It is a thin plastic or metallic circular plate coated with magnetic oxide and encased in a protective cover.

Data is stored on magnetic disks as magnetized spots.

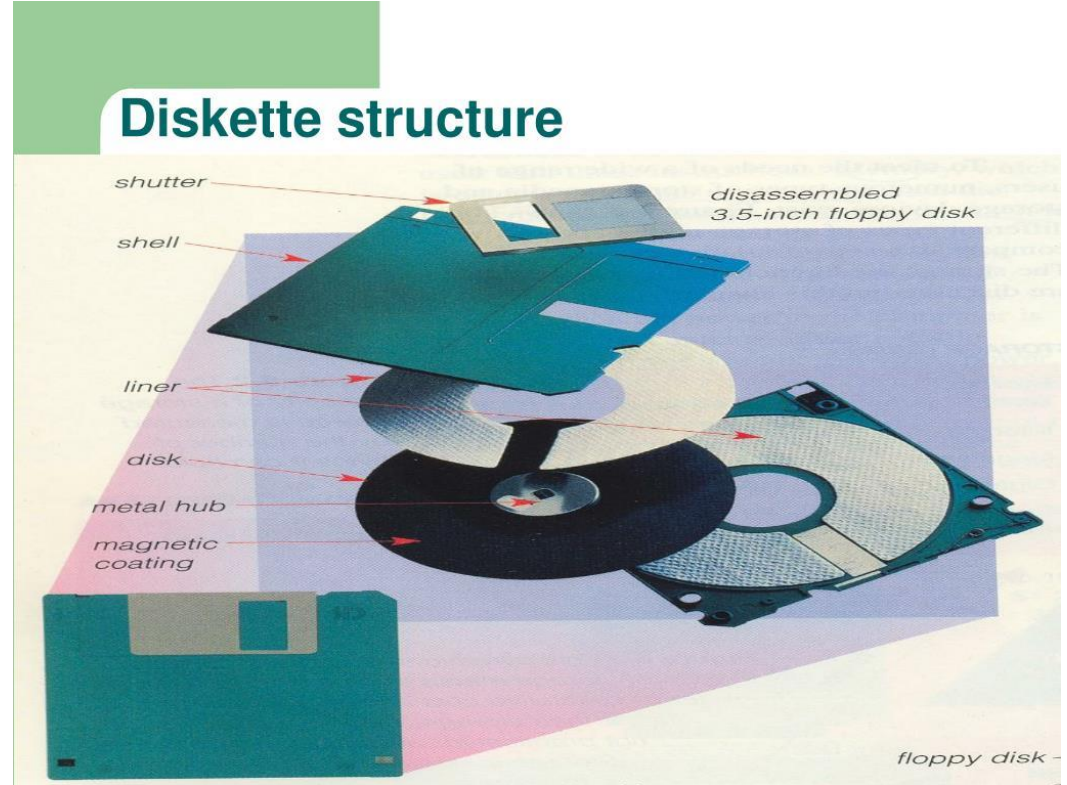
The presence of a magnetic spot represents the bit 1 and its absence represents the bit 0.

Eg: Floppy Disk and Hard Disk Drives

Floppy disk



most popular



Hard Disk

A Hard Disk Drive (HDD) is a storage device that **uses magnetic technology** to store and retrieve digital data.

It consists of spinning platters coated with a magnetic material and a read/write (R-W) head that moves to access data.

The disk is divided into tracks, and each track is further divided into sectors.

The outer tracks are larger than the inner tracks, they contain the same number of sectors.

Hard disk is a fixed disk. The disk is not removable from the drive, unlike floppy disk.

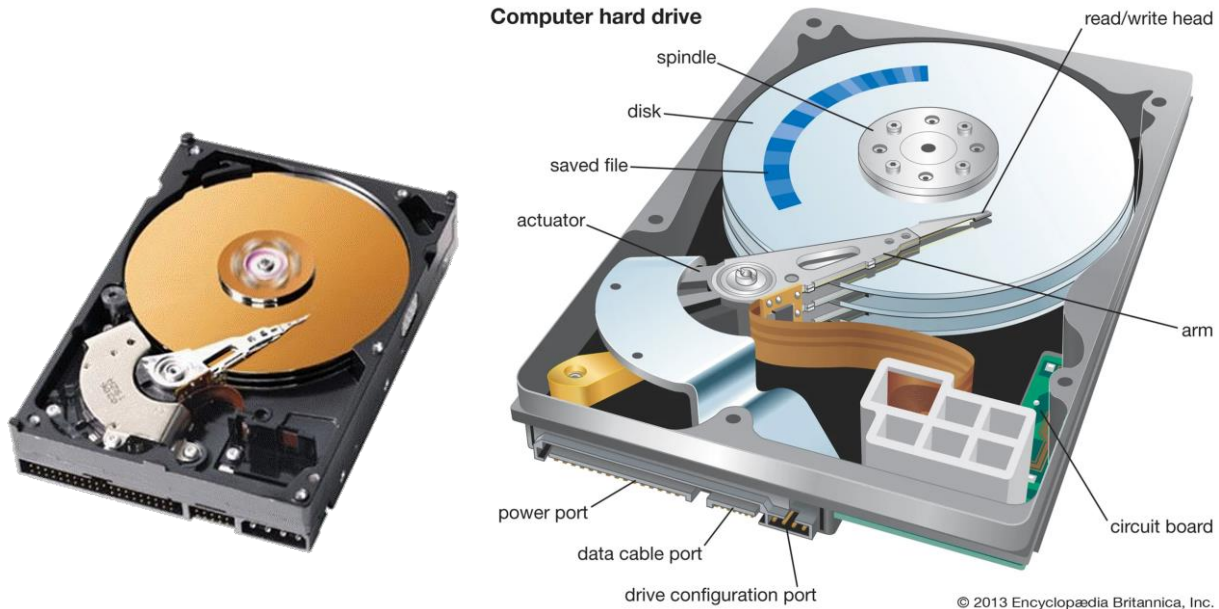
The hard disk and Hard Disk Drive (HDD) is a single unit.

Hard disk can store much more data than floppy disk.

The data in hard disk are packed more closely (because fast spinning uses smaller magnetic charges) and they have multiple platters, with data being stored on both sides of each platter.

Large capacity hard disks may have 12 or more platters.

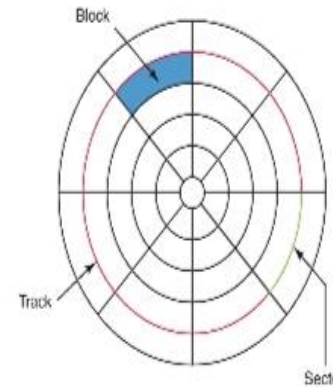
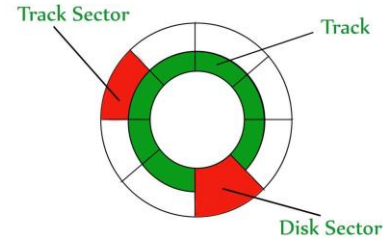
Unlike floppy disk, the read/write head of hard disk does not touch the disk during accessing.



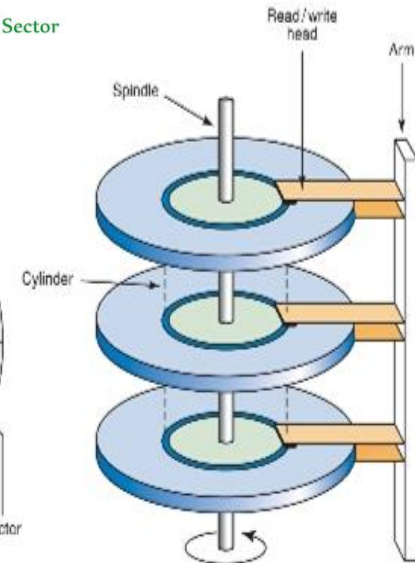
Components of Hard Disk

Structure of HDD

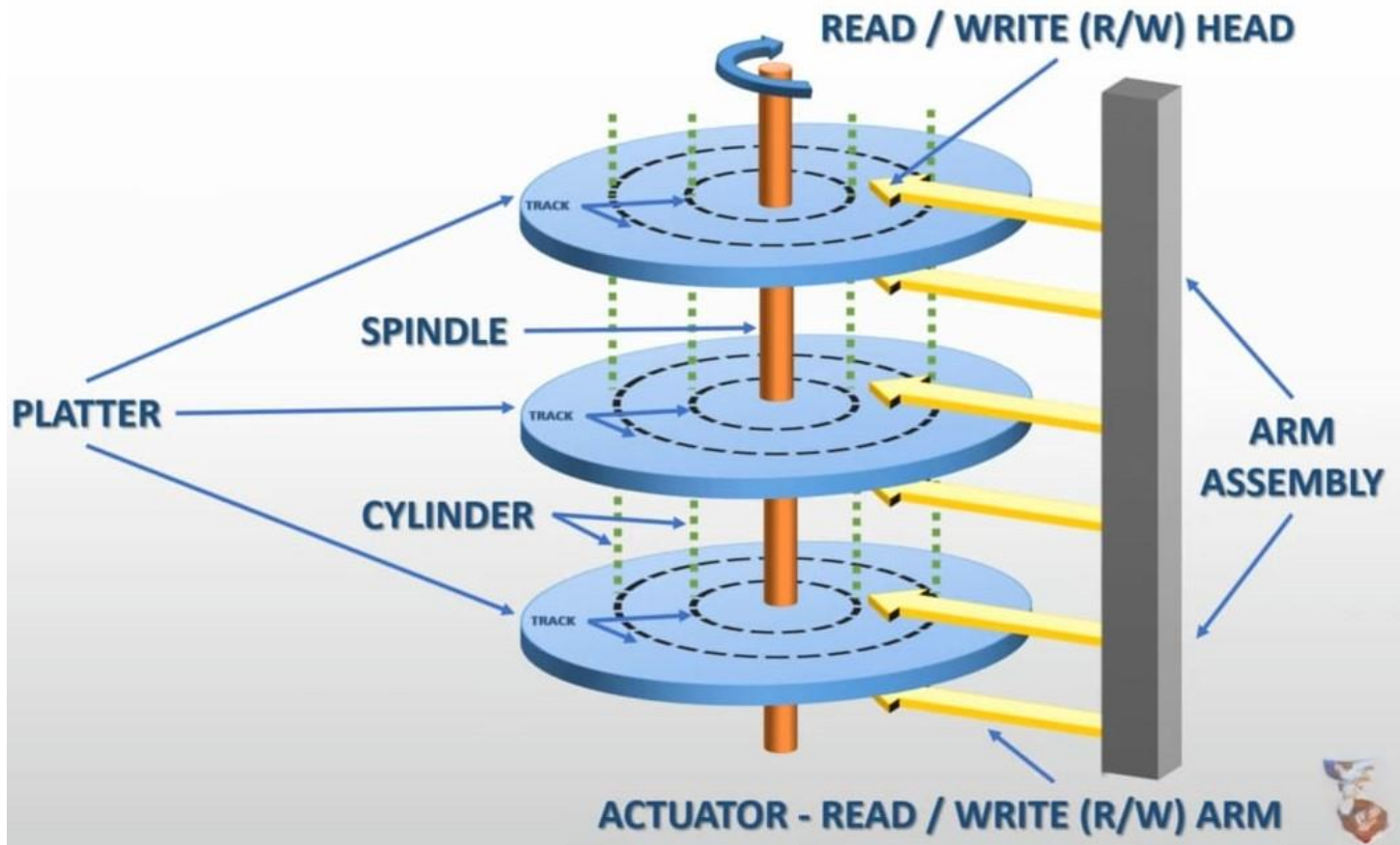
- The surface of disk is divided into concentric circles known as tracks.
- The outermost track is numbered 0 and the innermost track is the last track.
- Tracks are further divided into sectors.
- A sector is a pie slice that cuts across all tracks.
- The data on disk is stored in sector.
- Sector is the smallest unit that can be read or written on a disk.
- A disk has eight or more sectors per track

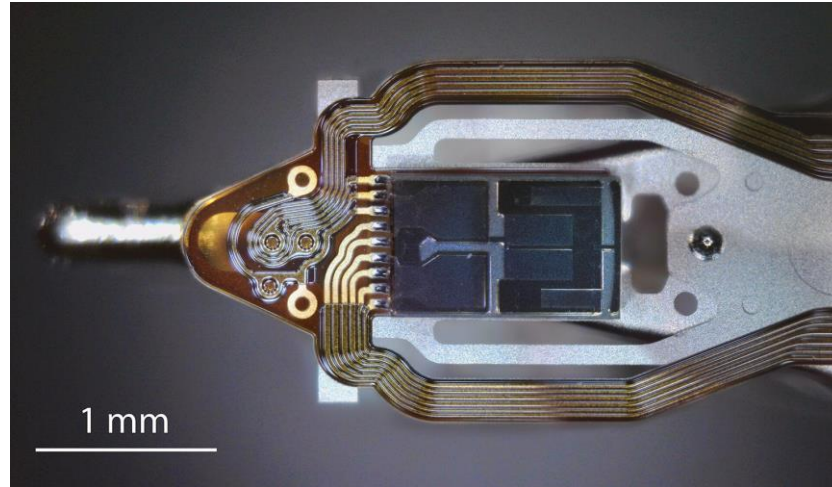


(a) A single disk



(b) A hard disk drive





Close-up of a single **read-write head**, showing the side facing the platter

It stores 512 bytes per sector but the number of sectors are more per track (54 or more) than floppy disk.

The operating system is stored on the hard disk.

The performance of computer like speed of computer boot up, loading of programs to primary memory, loading of large files like images, video, audio etc., is also dependent on the hard disk.

Magnetic disk is inserted into a magnetic disk drive for access.

The drive consists of a read/write head that is attached to a disk arm, which moves the head.

The disk arm can move inward and outward on the disk.

Components of HDD

Read/Write Head:

- The read head/write head is another mobile component of the hard drive.
- They are steered by an actuator that is driven by a second motor.
- Their pivoting motions allow them to sweep the entire surface of the drives.
- The role of the arm is to position the heads on a certain path to have access to the information.

Electronic component:

- Though the mechanical components are the most visible on a hard drive, the electronic component is just as important, because it handles the data transmission as well as procession and commands between the motherboard and the hard drive.
- Each hard drive has a microprocessor and an associative memory held on a printed circuit board (PCB).
- A signal processor handles the conversion of electric signals to digital signals.

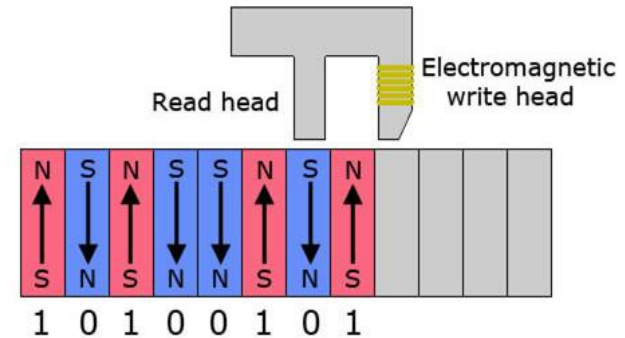
Working of HDD

Data is stored digitally in the form of tiny magnetized regions on the platter where each region represents a bit.

To write a data on the hard disk, a magnetic field is placed on the tiny field in one of these two polarities: North-South

An orientation in the one direction (like N-S) can represent the '1' while the opposite orientation (S-N) represents "0".

This polarity is sensed by integrated controllers built within the hard disk.



How Data is Stored and Retrieved

Writing Data

To write data, the HDD's controller moves the actuator arm to the correct track on the platter.

The read/write head then applies a magnetic field to the platter's surface, which changes the magnetic polarity of tiny areas.

These areas represent the binary data (1s and 0s) that the computer needs to store.

Reading Data

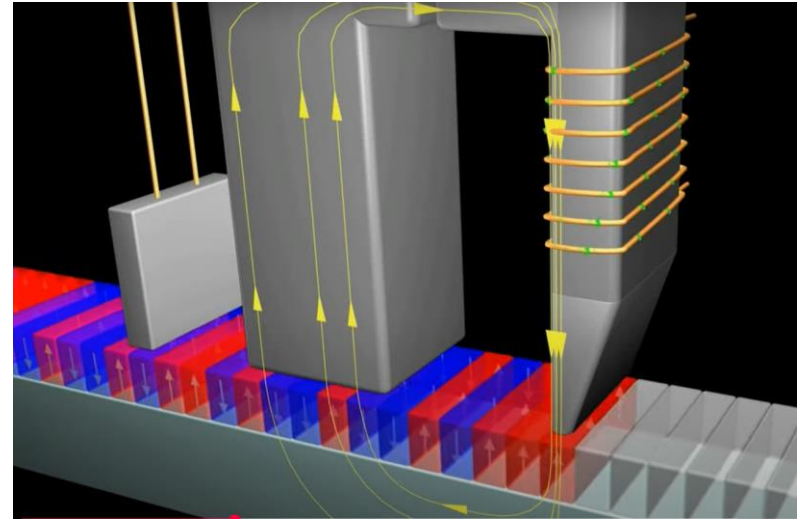
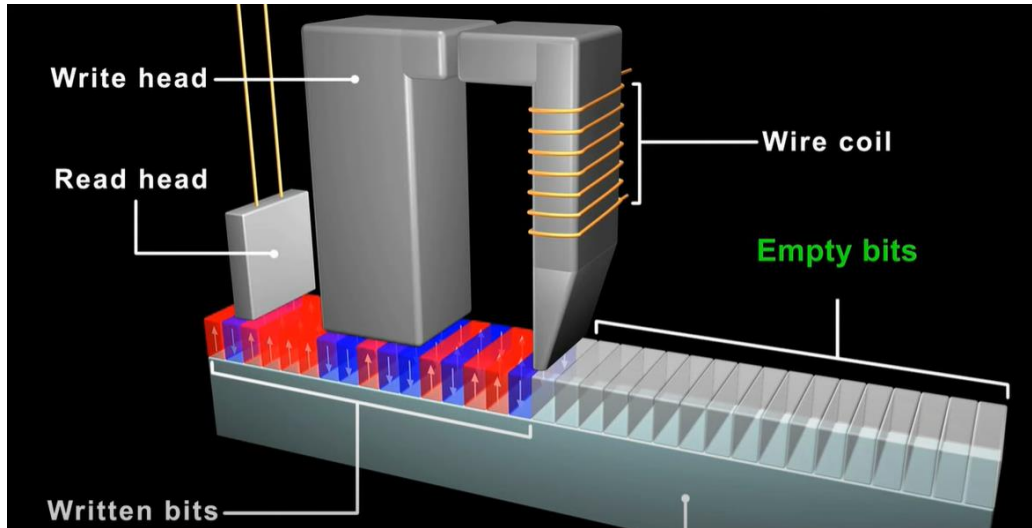
When reading data, the process is reversed.

The head moves to the correct track and senses the magnetic polarity of the areas on the platter.

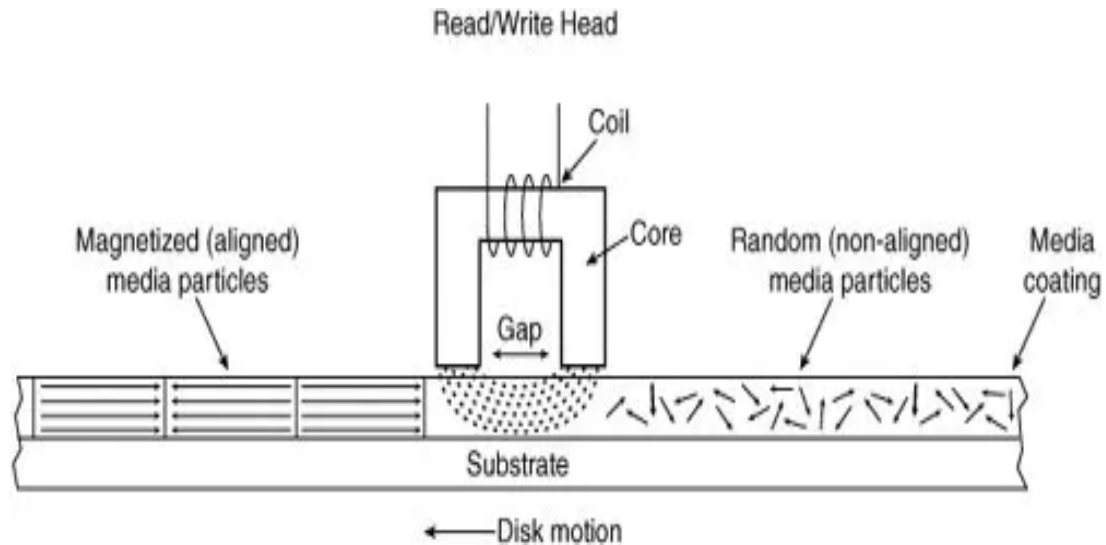
It converts these magnetic signals into an electrical current, which is then translated back into the binary data (1s and 0s) that the computer can understand.

Because the read/write head has to physically move across the spinning platters to find the data, HDDs are slower than SSDs, which use electronic flash memory with no moving parts.

Working of HDD



[Video Reference \(only for magnetism and HD part\):](https://youtu.be/f3BNHhfTsvk?si=46xuS2fg33CWnaly)
<https://youtu.be/f3BNHhfTsvk?si=46xuS2fg33CWnaly>



Block Diagram: Working of Magnetic Disk overview

Accessing data on the disk requires the following

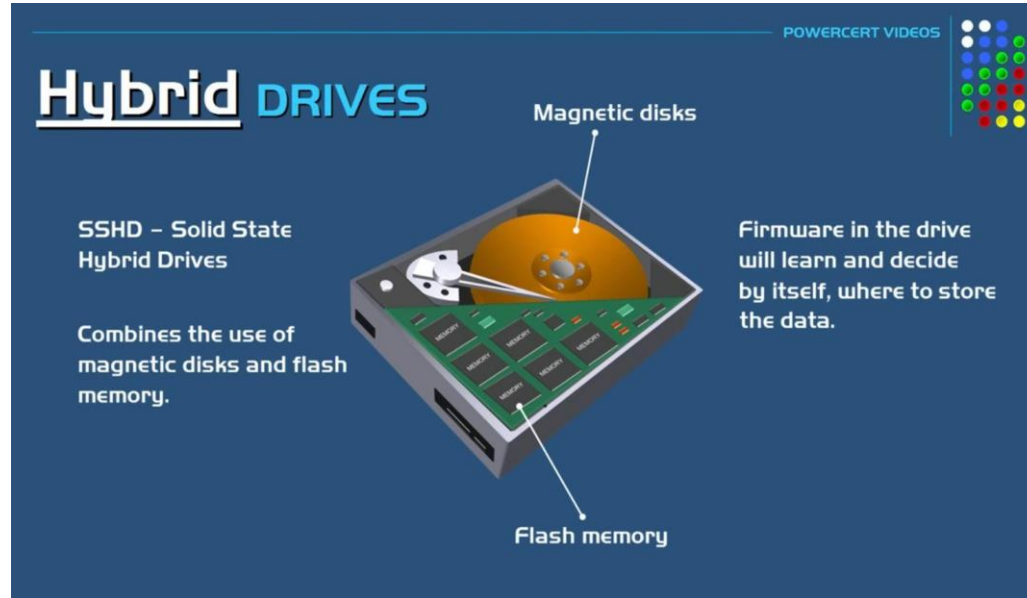
The read/write head is positioned to the desired track where the data is to be read from or written to. The time taken to move the read/write head to the desired track is called the **seek time**.

Once the read/write head is at the right track, then the head waits for right sector to come under it (disk is moving at high speed). The time taken for desired sector of the track to come under read/write head is called the **latency time**.

Once the read/write head is positioned at the right track and sector, the data has to be written to disk or read from disk. The rate at which data is written to disk or read from disk is called **data transfer rate**.

The **sum of seek time, latency time and time for data transfer is the access time of the disk**.

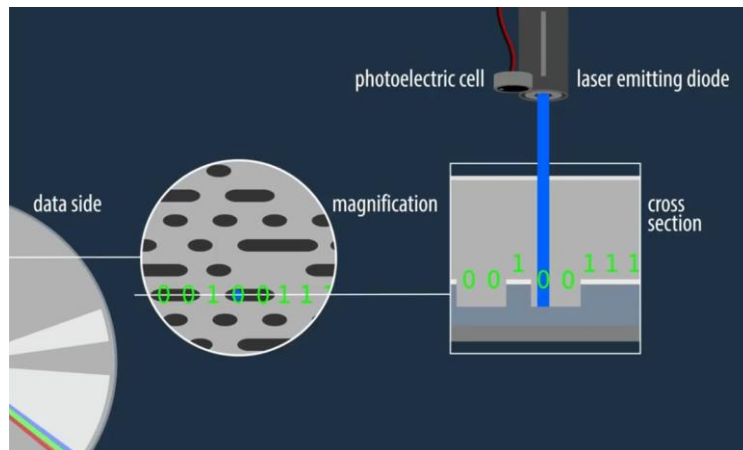
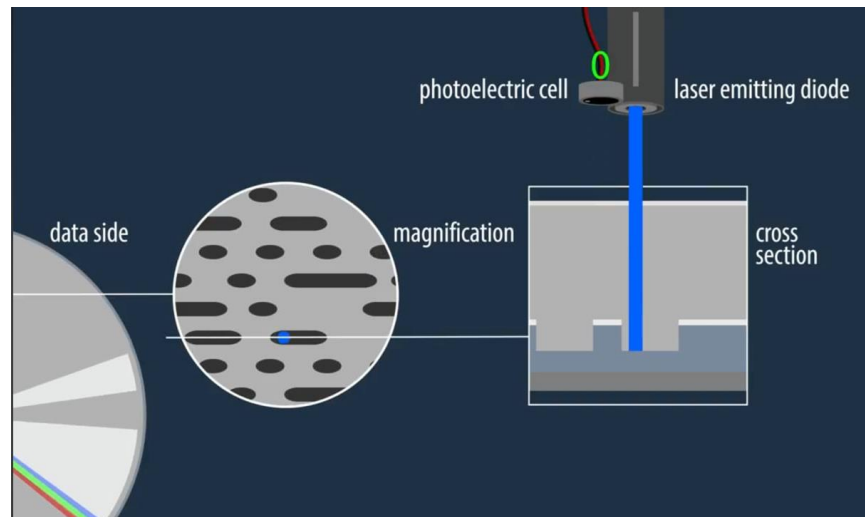
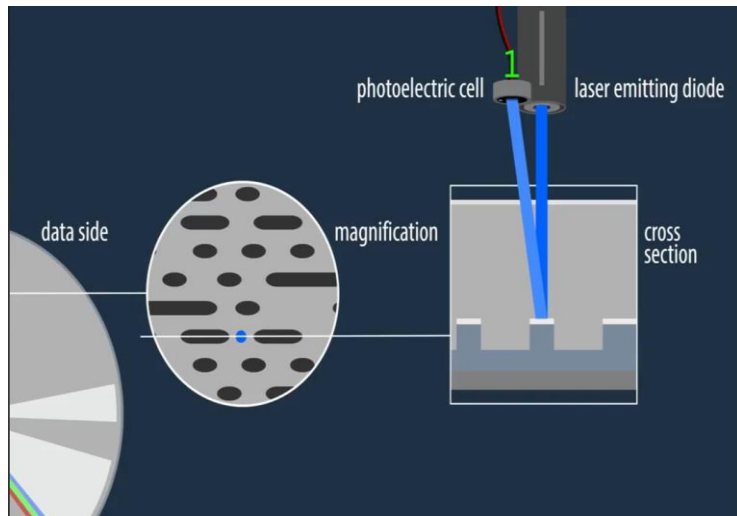
Hybrid Drives

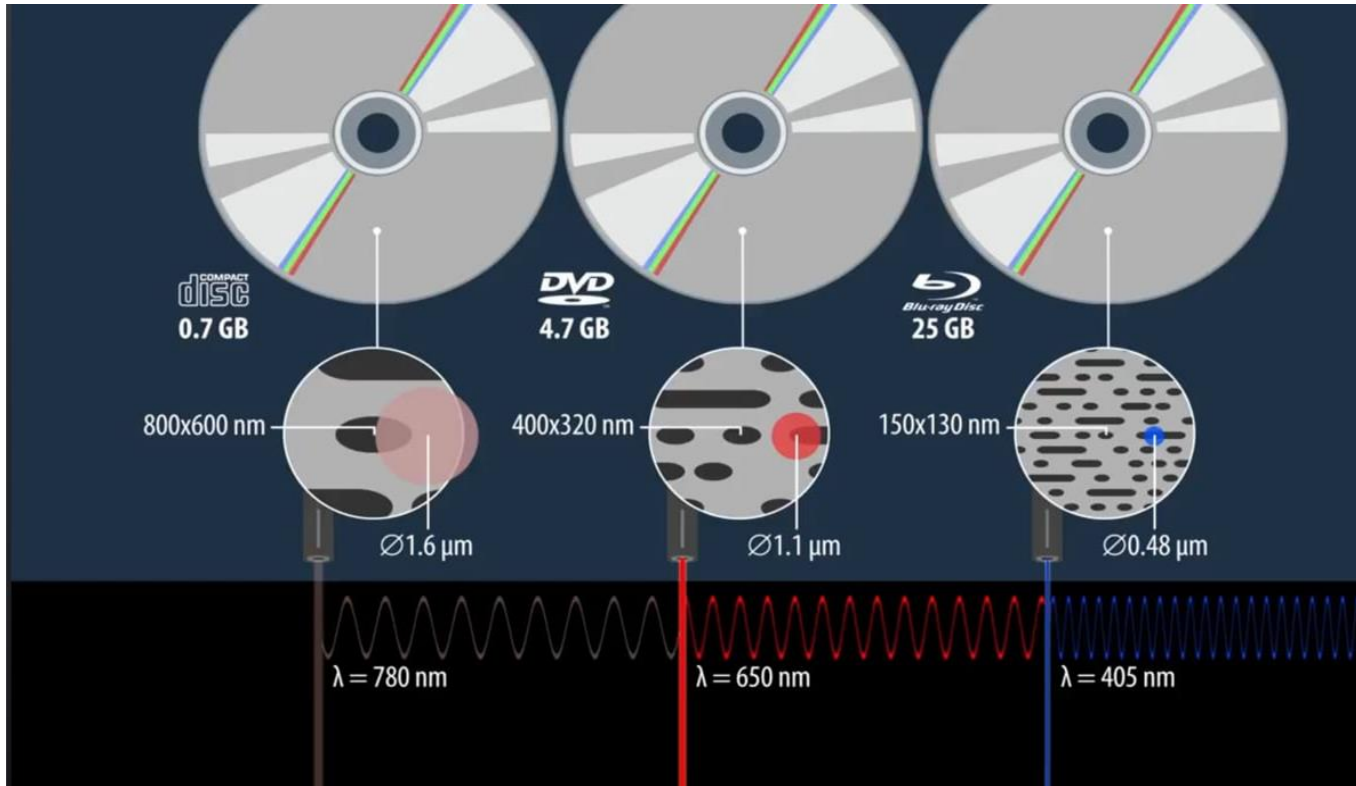


OPTICAL DISK

- Optical disk is a flat and circular disk which is coated with reflective plastic material that can be altered by laser light.
- Optical disk does not use magnetism.
- The bits 1 and 0 are stored as spots that are relatively bright and light, respectively.
- An optical disk consists of a single spiral track that starts from the edge to the centre of disk.
- Due to its spiral shape, it can access large amount of data sequentially, for example music and video.
- The random access on optical disk is slower than that of magnetic disk, due to its spiral shape.

- The tracks on optical disk are further divided into sectors which are of same length.
- Thus, the sectors near the centre of disk wrap around the disk longer than the sectors on the edges of disk.
- Reading the disk thus requires spinning the disk faster when reading near the centre and slower when reading near the edge of disk.
- Optical disks can store large amount of data, up to 6 GB, in a small space.
- Commonly used optical disks store 600–700 MB of data.
- The access time for an optical disk ranges from 100 to 200 ms.
- There are two most common categories of optical disks: read-only optical disks and recordable optical disks.





Solid State Drive

A Solid-State Drive (SSD) is a non-volatile storage device that stores data **without using any moving parts**, unlike traditional Hard Disk Drives (HDDs), which have spinning disks and mechanical read/write heads.

Because of this, SSDs are much faster, more durable, and quieter than HDDs.

They load files, programs, and operating systems quickly, making computers more responsive.

Since there are no moving parts, SSDs are also less likely to get damaged from drops or shocks.

Additionally, they use less power, which helps extend battery life in laptops.

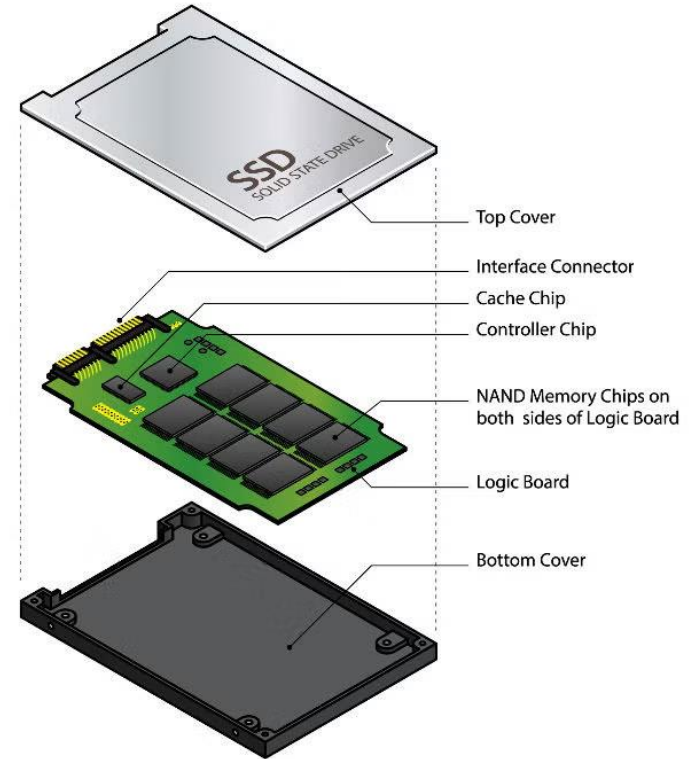
SSDs store data using non-volatile memory, meaning they retain information even when the power is off. T

his makes them a reliable and efficient choice for modern storage needs.

SSD structure and components

1. Outer Shell:

- The outer shell could be of metal or plastic and it helps in absorbing most of the heat from inside the flash memory.
- Since SSDs don't contain any moving parts, they give off very little heat and emit no noise.



2. NAND Flash:

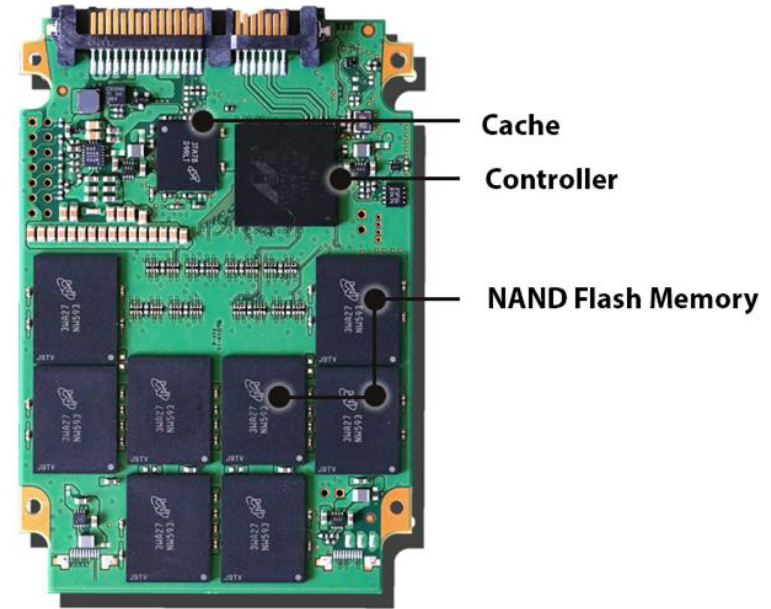
- Data is held in memory chips called NAND Flash.
- Because of this, the access time of SSD is really fast as the data is fetched directly from the NAND flash memory and the seek time is greatly reduced.

3. DRAM Buffer:

- A bit of DRAM is included in every SSD for the process of buffering.
- Similar to hard drive's cache, data is stored on it for some time temporarily before it is being written to the device.
- Wear-leveling data (Wear leveling is a process to distribute write/erase cycles evenly among all flash blocks of the SSD to prevent data from becoming unreliable and a particular block from wearing out) is also stored on the cache for some time during the running of the drive.

4. Controller:

The controller, normally a multicore-processor, is the heart of the SSD and it is the one that performs all the functions and is the main factor that separates one SSD from the other.



Working of SSD

Solid state drives (SSDs) use a combination of NAND flash memory technology and advanced controller algorithms.

NAND flash memory is the primary storage component, divided into blocks and pages.

An SSD contains a controller chip that manages data storage, retrieval, and optimization.

The controller's major duties are wear leveling, which evenly distributes write and erase cycles to extend the SSD's lifespan which consolidates empty blocks to maintain optimal performance.

SSDs also use features like the TRIM command to increase efficiency by telling the drive about wasted data blocks.

Furthermore, data compression and error correction techniques are used to increase storage capacity while maintaining data integrity.

NAND Flash Memory

NAND flash memory is a type of non-volatile storage used in SSDs, meaning it keeps data even when the power is off.

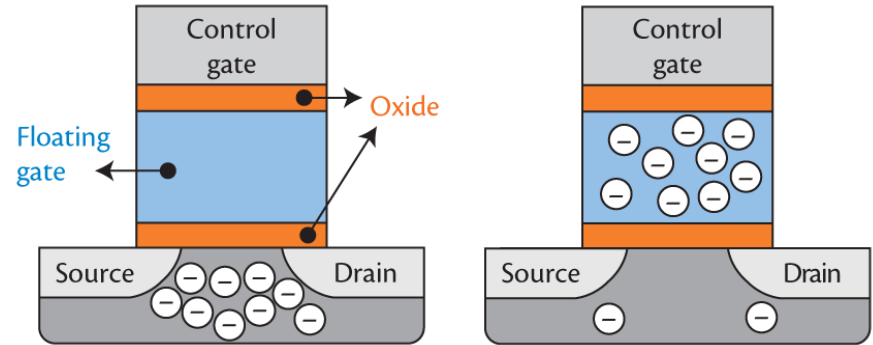
It gets its name from the NAND logic gate configuration of its internal circuitry.

The basic unit of NAND flash is a cell, which stores data as a charge on a component called a **floating gate**.

The NAND Cell: Each NAND cell is essentially a transistor with two gates:

- * Control Gate: This is connected to the rest of the circuit and is used to apply voltage to the cell.

- * Floating Gate: This is a small, insulated area that can trap and hold electrons. The presence or absence of a charge on this gate determines the cell's data value (a 0 or a 1).



MOSFET Transistor or Floating Gate (FG) MOSFET

Reading data from NAND cell

To read data from a NAND cell, a voltage is applied to the control gate.

The SSD's controller measures the electrical current flowing through the cell.

The amount of current is determined by the charge on the floating gate.

- * If there's a charge on the floating gate (trapped electrons), it restricts the flow of current. The controller interprets this as a binary 0.

- * If there is no charge on the floating gate, current flows freely. The controller interprets this as a binary 1.

This process is very fast and doesn't damage the cell.

Writing Data → more complex → "erase-before-write" process.

- * Read the Block: The SSD first reads all the data from the block that contains the page you want to update and temporarily stores it in its own memory (DRAM cache).

- * Modify Data: The new or updated data is combined with the old data from the block in the DRAM.

- * Erase the Block: The entire block is then erased, a process that clears all the charges from the floating gates and sets all the cells to a binary 1. This requires a high voltage that wears down the cell.

- * Write the Block: The SSD then writes the newly combined data (old data + new data) to an empty block in the memory.

This process involves applying a voltage to the control gate to force electrons into the floating gates, trapping them to represent the new data.

This erase-before-write limitation is the primary reason for a solid-state drive's limited lifespan.

Each erase cycle degrades the cell's floating gate, which is why NAND flash has a finite number of program/erase (P/E) cycles.

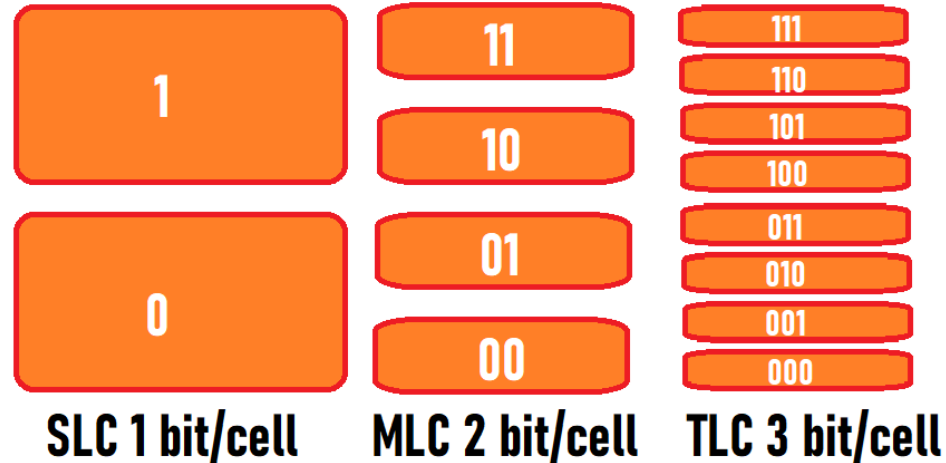
The SSD's controller uses a technique called wear leveling to spread write and erase cycles evenly across all the blocks, extending the drive's overall life.

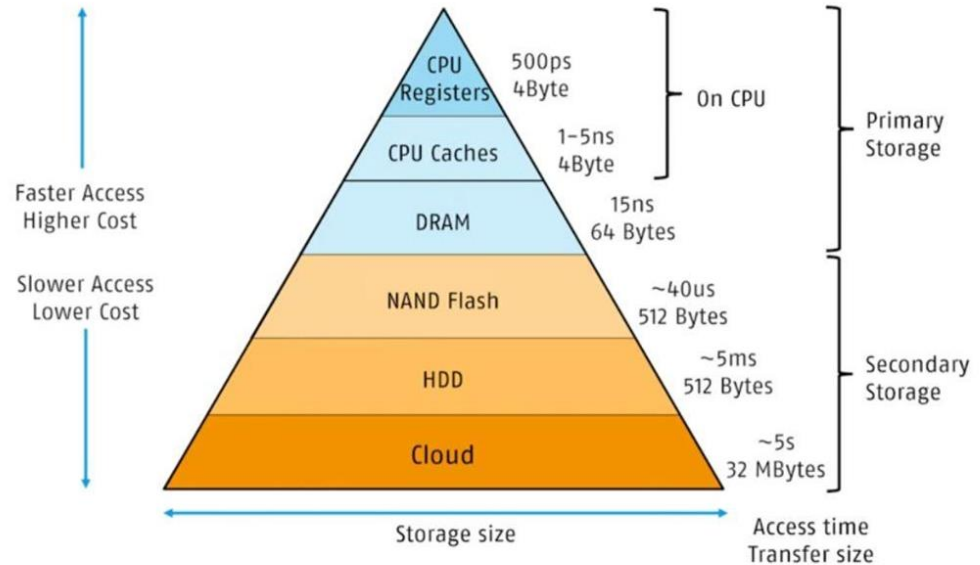
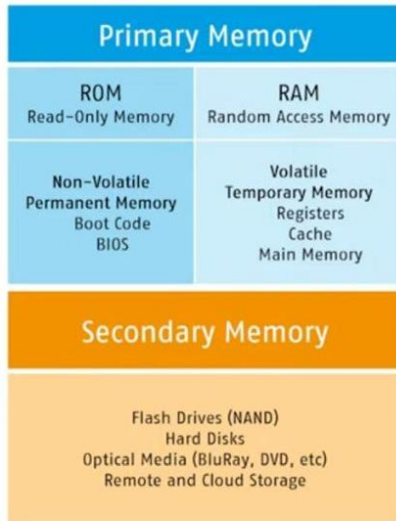
Wear leveling of SSD (just a visual representation)



Types of NAND (extra)

Types of NAND Flash Memory





Overall Memory Hierarchy